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Development of a Field Scale SWAT+ Modeling Framework for the Contiguous U.S.

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Research Impact Statement: The National Agroecosystems Model is a field based, national scale water quantity and quality model to aid in conservation planning and policy.

ABSTRACT: The Soil and Water Assessment Tool (SWAT) model is commonly used to predict the impacts of agricultural practices on water quality and quantity. Although widely applied, the data framework used to drive SWAT in the United States (U.S.) is fragmented and inconsistent, varying by user and model interface. This research describes the development of the National Agroecosystems Model (NAM), which provides a unified field to national scale modeling computational framework for research and decision support by using the latest SWAT platform, dubbed SWAT+. NAM has sufficient detail to capture field-level processes and management actions and spans the full extent of the contiguous U.S. NAM contains 7 million computational units, 4 million of which represent specific cultivated fields. It contains 3 million individually identifiable stream segments and more than 5,000 reservoirs. NAM is intended to serve as a reasonable base framework to be refined for specific applications. This work describes the individual data sources, assumptions, and the processing steps for their inclusion. NAM is constructed with 2,121 individual SWAT+ models which can be executed in a parallel hierarchical structure to dramatically improve runtime. This framework was tested in a case study of the Little River Watershed, Tifton, GA. NAM is developed using only publicly available data sources such that subsets of it can be shared to support research with other government agencies, universities, and others in the public domain.

(KEYWORDS: SWAT; simulation; non-point source pollution; rivers/streams; streamflow; surface water hydrology; spatially distributed modeling.)

INTRODUCTION

There is a strong need for scientifically credible information to guide agricultural and environmental decision-making at all levels of government. Federal agencies including the U.S. Department of Agriculture (USDA), U.S. Environmental Protection Agency (EPA), U.S. Geologic Survey (USGS), and National Oceanic and Atmospheric Administration (NOAA) make extensive use of sophisticated hydrologic models to inform policymakers on water-related issues. Agriculture is the largest consumptive user of water in the United States (U.S.), using twice as much as municipal, industrial, mining, and household usage combined (Dieter et al. 2018). Agriculture also is a major

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contributor to water quality impairment in the U.S. Runoff from agricultural fields may contain pesticides and excessive nutrients that fuel algal blooms or otherwise disrupt aquatic ecosystems. These water quality impairments are difficult to mitigate and severely limit the utility of water resources. Cultivated agriculture alone is responsible for 59% of nitrogen and 48% of the phosphorus entering the Gulf of Mexico via the Mississippi River (White et al. 2014). Given the interplay between water and agriculture, it is important that the data and models used to inform policy have a strong agricultural and scientific basis.

In recognition of this need, the USDA has long developed hydrologic models with an agricultural focus. One of the most widely used is the Soil and Water Assessment Tool (SWAT) (Arnold et al. 1998, 2012). SWAT is a distributed, process-based hydrologic and water quality model, most often applied at the watershed scale with a 30+ year history of application in national assessments. The entire contiguous U.S. was first simulated using SWAT by Srinivasan et al. (1998) as part of the Hydrologic Unit Model for the United States (HUMUS) effort. U.S. major river basins were later simulated as part of the Conservation Effects and Assessment Project (CEAP) (White et al. 2014). SWAT model development has continued with the recent release of SWAT+ (Bieger et al. 2017). SWAT+ features enhanced landscape and stream connectivity to better represent small watersheds and a consolidated file structure to facilitate application at regional or national scales.

Being relatively new, SWAT+ has only been applied to a few watersheds. Testing on the Little River Experimental Watershed in South-Central Georgia was conducted by Bieger et al. (2017) using an early version of SWAT+. Bieger et al. (2019) applied SWAT+ to the same watershed with the addition of landscape units and enhanced connectivity to test these routines. Yen et al. (2019) applied SWAT+ to the Middle Bosque River Watershed in the Texas-Gulf Region of Central Texas to test the Integrated Parameter Estimation and Uncertainty Analysis Tool (IPEAT+). The SWAT+ QGIS interface (https://swat.tamu.edu/software/plus/) was released in July 2019, which should increase the demonstrated usage of SWAT+ in future literature.

Within USDA, there are currently three efforts in need of a national-scale hydrologic/water quality model with an emphasis on agricultural management and effects: The Long-Term Agroecosystem Research (LTAR), the Agricultural Operations Planning Tool (AgOPT), and the continuing CEAP effort. CEAP began in 2003 as a response to an 80% increase in conservation spending associated with the 2002 Farm Bill (Mausbach and Dedrick 2004) and was designed to quantify the benefits of those increased expenditures. CEAP uses models to predict the fate and transport of sediment, nutrients, and pesticides from cultivated cropland with varying levels of conservation to U.S. waters. AgOPT is a tool under joint development by USDA-ARS, USDA-NRCS, and Texas A&M AgriLife to help producers schedule pesticide and fertilizer application to reduce losses in runoff. AgOPT uses a system of national models and precipitation forecasts to predict when runoff is likely to occur from cultivated fields. LTAR is a network of 18 established agricultural monitoring sites across the U.S. where data of uniformly high quality is collected (https://ltar.ars.usda.gov/). Both traditional and aspirational crop/conservation systems are examined at these sites. One objective of LTAR is to predict the nationwide environmental effects of increased adoption of these aspirational systems. Data collected at the LTAR sites will be used to calibrate models, which then can be applied to larger geographic areas. Despite their differing objectives, the needs of these USDA efforts can be addressed using a single model, provided it has sufficient detail to capture field-level processes and management actions, while also spanning the full extent of the contiguous U.S. (CONUS). Such a model is inherently large and complex and rarely undertaken.

Other SWAT models have been developed with similar extents, but not at the required level of detail. In SWAT and SWAT+, the smallest computation unit is the Hydrologic Response Unit (HRU); the number of HRUs is an indicator of the spatial granularity of the simulation. An HRU has specific land use, management, soil, climate, and topography. HRUs may be fields with distinct boundaries or an amalgam of non-contiguous discrete grid cells with similar characteristics. All previous large-scale efforts have utilized the latter approach because it is far less computationally intensive. Abbaspour et al. (2015) developed a model of Europe with 8,592 subbasins and 60,012 HRUs to inform water policy decision-making, which required a server-based network cluster for calibration. Schuol et al. (2008) simulated the entire continent of Africa in a single SWAT simulation with 1,496 subbasins and 1 HRU per subbasin. Climate changes across Iran were simulated by Abbaspour et al. (2009) using a SWAT model with 506 subbasins, also with 1 HRU each. Čerkasova et al. (2021) developed a transboundary Nemunas River model (Europe) using SWAT, comprising 11 interconnected sub-models, 9,012 subbasins, and 148,212 HRUs. This model is in use to guide the region's policy makers on climate change effects on the river and downstream lagoon environment, as well as possible nutrient reduction measures. Several of the very large river basins in China have also been simulated using SWAT including the lower Mekong (Mohammed et al. 2018), Yangtze (Sun et al. 2016), and Yellow rivers (Yu et al. 2017) with varying levels of spatial granularity.

SWAT models have been developed with individual fields represented as HRUs, but only for small

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watersheds. Merriman et al. (2018) developed a SWAT model of the Eagle Creek watershed (125 km^2) using 2,083 predefined field boundaries with very detailed management and conservation practices. A novel approach to include field boundaries in the ArcSWAT interface was developed and tested by Kalcic et al. (2015) in the little Pine Creek Watershed (56 km^2). They found field boundary-based and traditional HRU approaches yielded similar nutrient and sediment losses at the basin scale, but the field boundary-based models likely offered better understandability to producers and stakeholders. SWAT has also been applied where the entire model comprises only a single field (White et al. 2012) or multiple fields (Sommerlot et al. 2013). These applications demonstrate the applicability of SWAT at the field scale and the challenges of modeling at this level of granularity at a national scale.

These SWAT applications demonstrate that SWAT can operate at either the field or national scales, but the literature of multi-scale models is lacking. Such a model would be extremely large and complex, but there are no structural limitations in SWAT+ that precludes its development. The primary objective of this research is to develop a highly detailed national scale model to address the needs of the USDA. The National Agroecosystem Model (NAM) is currently under development by USDA and Texas A&M AgriLife Research and Extension for this purpose. The development of NAM model components is described herein. NAM covers the entire contiguous U.S. and selected drainages of Canada and Mexico. This region is extremely diverse in terms of geology, climate, and agriculture, making it an excellent testing platform for SWAT+. A secondary objective of this research is to test the NAM framework by application to a research watershed in the LTAR network.

METHODS

All datasets used in the development of NAM are national scale and derived from publicly available sources. SWAT+ requires a variety of spatial and non-spatial data. Due to the unique nature and requirements of NAM, all data processing was conducted with custom tools instead of the ArcSWAT or QSWAT interfaces.

Discretization and HRUs

The NAM SWAT+ model was constructed using 86,016 subbasins taken directly from Hydrologic Unit Code 12 (HUC12) boundaries as defined in the 2015 Watershed Boundary Dataset (WBD) (USGS and USDA/NRCS 2013) These data were subject to extensive manual attribute correction for downstream flow connections between HUC12s (White, Beiger, et al. 2017). Each HUC12 was further subdivided into HRUs. The vast majority of cultivated cropland HRUs were based on predefined field boundaries and have contiguous areas. The remaining areas were represented using a more traditional HRU approach where unique combinations of soil and land use are defined, and non-contiguous areas are grouped and filtered into HRUs.

Cultivated cropland field boundaries were defined for the US using data published by Yan and Roy (2016), who delineated 4.1 million fields using a combination of weekly 30 m Web Enabled Landsat, (WELD), Landsat 5 Thematic Mapper (TM) and Landsat 7 Enhance Thematic Mapper Plus (ETM+) imagery. While it would be preferable to use field boundaries derived from higher resolution imagery, such as those manually digitized for USDA's Common Land Unit (CLU), these are not publicly available. Manual examination of fields by Yan and Roy (2016) in agricultural areas revealed the most common error to be an excessive subdivision of fields, which should have minimal impact on SWAT+ model predictions. Raster field boundaries by Yan and Roy (2016) were vectorized and simplified prior to overlaying with soils, topography, and land use data. After processing these field boundaries tended to be slightly inset from boundaries visible in aerial photography. This slightly reduced the area of each field and tended to exclude areas adjacent to fields such as streams and road ditches from each field's polygon. Stream and road ditches have much different soil and topographical characteristics, thus excluding these likely leads to more appropriate parameterization of HRUs representing these fields. Adjustments to ensure agreement with total land use areas at the HUC12 level as provided by the Cropland Data Layer (CDL) (Han et al. 2012) were applied later.

Field boundaries were overlain with a 10 m Digital Elevation Model (DEM) from the USGS National Elevation Dataset (Gesch et al. 2018) to determine field slope. The dominant soil was taken from 10 m Gridded Soil Survey Geographic (gSSURGO) (Soil Survey Staff 2014) database using a process detailed by Di Luzio et al. (2017) and Di Luzio (2017). Where these data were not available, U.S. General Soil Map (STATSGO2) (Soil Survey Staff 2021) was used. The full set of SWAT+ soil parameters by layer was utilized. Other SSURGO data such as minimum, maximum, and typical values for slope and slope length were used to corroborate topographical parameters derived from DEMs.

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Land use for each field was derived from the CDL (Han et al. 2012) with crop rotation information derived from CDL data from 2013, 2014, 2015, and 2016. CDL data are reasonable reliable source for these rotations with a class-specific accuracy for all crop classes of 92% in 2016 (Lark et al. 2021). CDL data were reclassified from 133 categories to 43 by incorporating minor categories with larger similar ones (e.g., sweet corn, ornamental corn, and popcorn to field corn) or crops of a particular class (e.g., chickpeas, lentils, peas, to legume row crops). Major field crops, forests, and urban areas were not reclassified.

Non-cultivated land uses and cropland where field boundaries were not available were represented using a more traditional HRU approach. For each of the 86,016 HUC12s, available soils (SSURGO or STATSGO2) were overlaid with 2014 CDL land use. Unique combinations were identified and intersected with the 10 m DEM to estimate a slope for each potential HRU. This overlay encompassed the entire HUC12, including areas with field boundaries, thus field based HRUs are also represented in this collection. A total of 26,320,923 potential HRUs were identified; therefore, the resulting model is far too complex and the run-time too long for any practical use. Thus a set of rules were developed to aggregate these HRUs and combine them with the field-based HRUs, while preserving the most important aspects of the overall composition within each HUC12.

1. The overall land use fractions at the HUC12 level from the CDL are preserved, and data derived from these overlays are considered authoritative. This may result in the grouping of HRUs with the same land use and differing soil types. If both soil and land use were preserved, many times more HRUs would be necessary. They are correlated as land use suitability is largely determined by soil characteristics and topography.
2. Potential HRUs that match the characteristics of an existing field-based HRU are considered duplicative and removed. A hierarchical routine uses soils and land use to identify matches while tracking the overall cumulative areas. The area of each field-based HRU can vary within a predefined range to account for the previously mentioned in-setting and delineation errors.
3. Open water is excluded. A properly configured SWAT+ model should not include water as an HRU. Water areas are represented by ponds, reservoirs, or channels.
4. Each HRU has a maximum potential size based on its land use type. This prevents large portions of a HUC12 from being subject to an identical management schedule. The maximum size ranges from 2 km^2 for managed grasslands and crops to 10 km^2 for rangeland and forests.
5. At least one HRU of each land use type present in a HUC12 is included. This prevents the loss of relatively small yet important HRUs (e.g., a small urban area in a largely forested HUC12).
6. If no field boundaries exist in a HUC12, and it contains cultivated cropland, NAM uses traditional HRUs.
7. A single predefined field that crosses a HUC12 boundary is represented as two separate HRUs with identical management.

Due to the size of these data, they are stored and manipulated in Microsoft SQL Server tables. Microsoft. NET was used to develop software that interacts with the SQL (Structured Query Language) server and processes the data. After processing, NAM was divided into 4,606,927 field-based and 2,296,974 traditional HRUs (6,903,901 total). The average field HRU size is 24 ha, while non-field HRUs average 323 ha. The mixing of field and non-field HRUs does have the potential to introduce additional uncertainty. A new HRU configuration for NAM is under development where the entire landscape is fully discretized into contiguous elements using neural networks algorithms and aerial imagery (Figure 1).

Management

Each HRU in SWAT+ has a specific set of management operations including tillage, planting, fertilization, irrigation, and harvest. Structural conservation practices, tile drainage, and tillage intensity are also required. These SWAT+ data are among the most difficult to obtain, and many large-scale modeling efforts utilize default management schedules based on heat unit scheduling. Given the purpose and multi-scale application of NAM, we required much more specific data.

Irrigation. Irrigation is an important part of the water balance and must be considered in NAM to obtain reasonable results. Irrigation is a common practice in arid and semiarid regions and may be used in more humid areas for specialty crops, managed grasses, and periodic drought mitigation. Irrigation is also limited to areas with adequate ground or surface water supplies and varies regionally even in areas with similar climates. These factors make it difficult to predict which HRUs utilize irrigation.

Several data sources were used to develop the NAM irrigation data. County/agricultural district level summaries of irrigated and non-irrigated planted acres were obtained from the National

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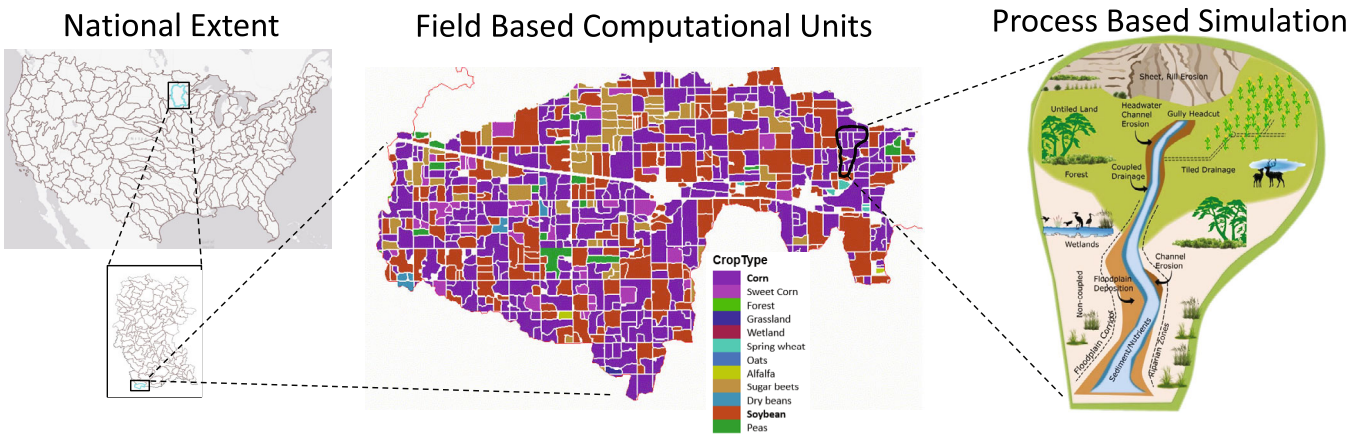


FIGURE 1. National Agroecosystems Model (NAM) scope and scale.

Agricultural Statistics Service (NASS) for the years 2010 to 2018. These data were summarized by crop type (corn, wheat, cotton, oats, soybeans, barley, alfalfa, non-legume hay, and sorghum). For each county where sufficient data were available, a probability of irrigation was estimated (fraction of irrigated acres/total acres). Unfortunately, many counties lacked data on one or more crops known to grow there. To construct a predictive model to fill these missing values, other county-level data were assembled including average precipitation, crop yields, median Potential Evapotranspiration (PET), and irrigation withdrawals. Various regression and machine learning techniques were attempted and evaluated. For the purposes of transparency, a relatively simple regression model with some knowledge-based limits was employed. Where data were available for multiple crops, irrigation probabilities were highly correlated. For example, 89% of the variance in cotton irrigation could be explained using the average irrigation probability across all crops excluding cotton, and according to these data, cotton fields are 1.16 times more likely to be irrigated than the average crop. Similar relationships were developed for other crops. A minimum precipitation threshold value below which all HRUs are irrigated was also established for each crop. These thresholds were based on irrigation probability vs. precipitation regressions, comparisons of reported high crop yields in counties with low precipitation, and professional judgment. Each HRU was assigned an irrigation status based on this database of probabilities. Minor crops not available in the database were assigned either a surrogate, the average, or the maximum probability of all crops.

Pasture and hay land may be irrigated, which increases production, stocking rates, fertilizer demand, and cutting frequency. The probability that a particular HRU was irrigated was defined by NASS county-level summaries of irrigated and non-irrigated planted acres. HRUs assigned as irrigated utilized modified management operations also based on expert opinion. Automatic irrigation was assigned using SWAT+ decision tables. Rice and urban areas were assumed always irrigated. Urban areas where rainfall is high would receive little irrigation because it is demand based.

Management Operations. Non-Cultivated. Management for non-cultivated HRUs at the HUC8 level was derived from an expert panel of simulation modelers and USDA Natural Resources Conservation Service personnel. This same management was also implemented in the first generation of CEAP. Forest and brushy rangeland are established and subject to no further operations. Urban (the pervious portion) is established with a warm or cool season grass depending on latitude and subject to mowing and two fertilizations per year. Grassy rangeland is established and grazed at a low stocking rate. Pasture is established with a warm or cool season grass and grazed based on the expert panel recommended stocking rates. Hay land may be alfalfa, fescue, or Bermuda and is cut periodically and fertilized to maintain productivity. Pastures and hay land may also receive additional manure based on agricultural census data analyzed by Gollehon et al. (2016).

Cultivated. Management for cultivated cropland was defined based on management templates provided by White et al. (2016). These data were derived from NRCS crop templates for the RUSLE2 model (Foster et al. 2001), US agricultural census fertilizer application data, and NASS county-level yields for major field crops. These data were processed into 20,000+ SWAT compatible management templates describing a wide range of crops, two-year crop

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rotations, and tillage schemes for each of 72 NRCS Crop Management Zones (CMZ). For each HRU in NAM, a single template from this database was selected based on a variety of criteria.

For each field-based HRU, multiyear land use (2014, 2015, and 2016 CDL) was used to identify the primary crop or first two-year crop rotation if present. These data were used to identify a subset of suitable management templates within or adjacent to the CMZ in which the HRU was located. If a suitable rotation could not be located, a matching single crop template was selected.

The resulting subset was further narrowed by tillage intensity. Baker (2011) provided tillage intensities by crop type and HUC8, which were derived from field surveys collected by the Conservation Technology Information Center. Baker (2011) contains the fraction of cropland in one of five tillage categories. This fraction was used to define the probability that an HRU would have a particular tillage class, and the assignment was made. The assigned tillage class was used to define a range of acceptable Soil Tillage Intensity Rating (STIR) values, which in turn were used to further reduce the subset of matching management templates for the HRU in question. A single template was selected at random from the remaining subset to represent the HRU.

Most management templates from White et al. (2016) include the use and rates for commercial fertilizers, but manure applications are also common in regions with Concentrated Animal feeding Operations (CAFOs). At our request Gollehon et al. (2016) prepared additional data tables of cropland manure application summarized at the HUC6 level. These data included a fraction of cropland acres receiving manure and total nitrogen and phosphorus application. The fraction was used to define the probability that each HRU received manure. If selected an application of manure with the equivalent nutrient content was applied.

Tile Drains. Tile drainage was simulated in selected cultivated fields based on distributional data derived from Sugg (2007) and Finocchiaro and Tangen (2014) then summarized by Schwartz (2015). These data provide county-level estimates of cropland under tile drainage. These data were redistributed to the HUC8 level based on area-weighted approach. Both the original tile data and the redistribution method contain a great deal of uncertainty. There are many undocumented old tile systems in the Midwest that still function to some degree.

Software was developed to rank all cropland HRUs within each HUC8 according to soil characteristics associated with a need for tile drainage. These included poorly drained soils with hydrologic soil group D or differing groups for drained vs. non-drained conditions and field slope. Within each HUC8, ranked HRUs were sequentially tagged as tiled until the total area of tiled HRUs matched the HUC8 estimate. Initial testing of this procedure identified fields that clearly should have been tiled that were not and an overall underestimation of tiles of about 15%. SWAT+ simulations indicated severe crop yield underpredictions in these fields due to aeration stress. We partially attribute this to the transformation of county-level data to a HUC8 basis and underreported tiles. This transformation assumed a uniform distribution of tiles within each county, which is unlikely. Discussions with tile drainage experts in the Midwest confirmed the likely underreporting of tiles. We added an additional allowance to exceed our HUC8 estimates if the field would require tiles to be in production. Our final tiled area in the NAM model was 16% greater than the area estimated by Schwartz (2015).

Structural Conservation. Data for structural conservation practices at the national scale are limited. CEAP conducted extensive (18,000+ points) surveys for 2003–2006 and 2015–2016, but these data are proprietary and are not publicly available. Various state NRCS agencies also maintain contract records of cost-share programs that establish structural conservation, but the availability of these varies, and they do not include all practices on the landscape. For NAM we elected to derive structural conservation data for terraces, waterways, filter strips, riparian buffers, contour farming, and strip cropping from White, Haglund, et al. (2017). Their data were based on a survey conducted using publicly available aerial photography and contain digitized field boundaries and Boolean attributes for each structural conservation practice. We overlaid these boundaries with soils, land use, DEM, and HUC8 watersheds. A database of 8,079 cultivated fields was developed with soil properties, field slope, structural conservation, and HUC8.

Our approach to applying these data to NAM HRUs was to identify a nearby sample with similar characteristics and apply all structural conservation from that sample to the HRU. Structural conservation practices are not independently distributed. Conservation planners design a system of multiple practices that work together, thus it is important that the entire conservation system is intact in NAM. For each cultivated HRU in NAM, the conservation database was searched for samples with a similar slope and hydrologic soil group that also resides within the same HUC8. If three or more matching samples were found, one was selected at random, and all practices in that conservation system were applied to the

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HRU. If less than three were identified in the HUC8, the search was repeated at the HUC6 level. If necessary, the search was repeated at increasing levels (HUC4, HUC2, entire U.S.) until at least three matching samples were identified and one was selected at random and applied to the HRU.

Weather

One set of observed daily weather data is used for each HUC12 in NAM. These daily data include precipitation and maximum and minimum temperatures. Early versions of NAM relied entirely upon interpolated and processed data from the Global Historical Climatology Network (GHCN) as described by White, Gambone, et al. (2017). They documented potential issues using these data for hydrologic modeling in the western U.S. due to orthographic effects and lower station density. To mitigate these issues, climate for HUC2 regions 13, 14, 15, 16, 17, and 18 were also derived from Parameter-elevation Regressions on Independent Slopes Model (PRISM). PRISM is adjusted for orthographic effects common in mountainous regions and has been shown to have higher accuracy in small watersheds (Gao 2017). PRISM data were processed using methods described by Gao et al. (2020).

Streams and Rivers

Stream and rivers were defined in NAM using the National Hydrography Dataset Plus Version 2 (NHDPlus) (Moore and Dewald 2016). Earlier versions of NAM included only main channel segments, one for each HUC12. White, Beiger, et al. (2017) described the processing and simplification of NHDPlus for this purpose. The version of NAM presented here attempts to utilize all stream segments within the NHDPlus. It is comprised of more than 3 million individual channel segments derived from "blue line" streams in 1:100,000 USGS topographic maps. These data have been processed to include additional attribute data useful for inclusion in hydrologic models. Channel width and depth were estimated using relationships developed by Bieger et al. (2016). In HUC12s where NHDPlus was not available or otherwise unusable, regional (HUC8, HUC6, or HUC4) average main and tributary channel properties summarized from White, Beiger, et al. (2017) were used.

Streambank and bed characteristics have a tremendous influence on sediment and nutrient transport, but only a small number of U.S. stream segments have measured data. We overlayed gSSURGO with each NHDPlus segment to estimate stream bank soil texture. Each segment was also intersected with the CDL to identify bank vegetative class (cropland, grassland, urban, bare, forest). Generalized bank vegetation classes were assigned a cover factor ranging from 0.1 (bare) to 0.75 (forest) based on erosion resistance. The channel cover factor for each channel segment was taken as a weighted average.

The median particle size of bed material (D50) is a critically important factor in sediment transport. These data are available for a very limited number of NHDPlus channel segments, however, Asamen et al. (2022) summarized and analyzed 2,208 observations of D50 collected across five different surveys. These data were used to develop relationships between D50, drainage area, and channel slope for five regions (Northeast, Midwest, Southeast, Southwest, and Pacific). These relationships were used to estimate D50 for all channel segments in the NHDPlus network.

Reservoirs, Ponds, and Wetlands

In SWAT+ reservoirs and ponds are simulated as separate spatial objects independent of the subbasin, while wetlands are treated as HRUs, in which water can pool. We define ponds as impoundments that receive surface runoff from other HRUs but are not directly connected to the stream network as defined by the NHDPlus. Ponds within a HUC12 are simulated collectively as a single structure to reduce complexity. All HRUs within a subbasin contribute a fraction of their runoff volume to the pond based on the assumed drainage area required to support that pond. Ponds are considered part of the upland landscape water balance.

Major reservoirs are large structures attached to a main channel within the stream network. These include managed reservoirs used for water supply or flood control. Minor reservoirs are smaller unmanaged structures attached only to a tributary channel such as those associated with the Watershed Protection and Flood Prevention Act (PL-566). Half of all HUC12s contain a reservoir of some type. Reservoir release parameters were derived for broad categories using data presented by Wu et al. (2020).

Wetlands are defined by the 2011 National Land Cover Database (NLCD) (Homer et al. 2012), which also serves as the basis for the annual CDL release. Wetlands are the only HRUs that receive runoff from other HRUs. Like other non-contiguous HRUs wetlands are consolidated, but there may be multiple wetlands in each HUC12 provided they have different soils. As HRUs, wetlands are part of the upland water balance.

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Connectivity

Previous SWAT versions utilized a rather idealistic configuration of objects within a subbasin. All HRUs, are grouped into a single subbasin, which is in turn partially routed through an optional pond or wetland to a single main channel. This main channel transfers flow to an optional single reservoir and then to the subbasin outlet (Figure 2). Shallow and deep aquifers are present in each subbasin. All surface and subsurface flows are eventually routed to the subbasin outlet. Non-contributing areas or unconnected subbasins are not permitted.

An examination of the connectivity between impoundments, streams, and wetlands, both within and between HUC12 was conducted to identify the potential range in real-world conditions across the CONUS. Some examples are given in Figure 2. It is important that the connectivity in NAM be flexible for a wide range of conditions.

Connectivity between objects is extremely flexible in SWAT+. This allows a wide variety of complex hydrologic systems to be represented, but there are drawbacks. It is possible to configure object connectivity such that "infinite" loops are constructed, which prevent SWAT+ from running properly. Arnold et al. (2021) described a conceptual connectivity framework for NAM. Since transition channels were not available for every field, the conceptual framework was simplified within a HUC12. In NAM, we group all non-wetland HRUs within a HUC12 into routing unit #1, which is effectively a container object to simplify routing and reporting. If wetlands and ponds are present in the HUC12, a portion of the flow from routing unit #1 is directed to these objects and the remainder is routed directly to routing unit #2 which serves as an object to store all surface flow prior to routing through streams. Routing unit #2 also provides an opportunity to summarize and report all upland loads before routing. After being subjected to relevant processes, flow from wetlands and ponds is also directed to routing unit #2. Subsurface flows below the soil profile from all HRUs, wetlands, and ponds are directed to a single HUC12 shallow aquifer.

Flows from routing unit #2 (surface) and the HUC12 Aquifer (groundwater) are distributed among all NHDPlus channel segments based on incremental drainage area (drainage area contributing to that segment only). Incremental drainage areas for channels that intersect reservoirs are attributed to that reservoir, and appropriate flow is directed there. Only

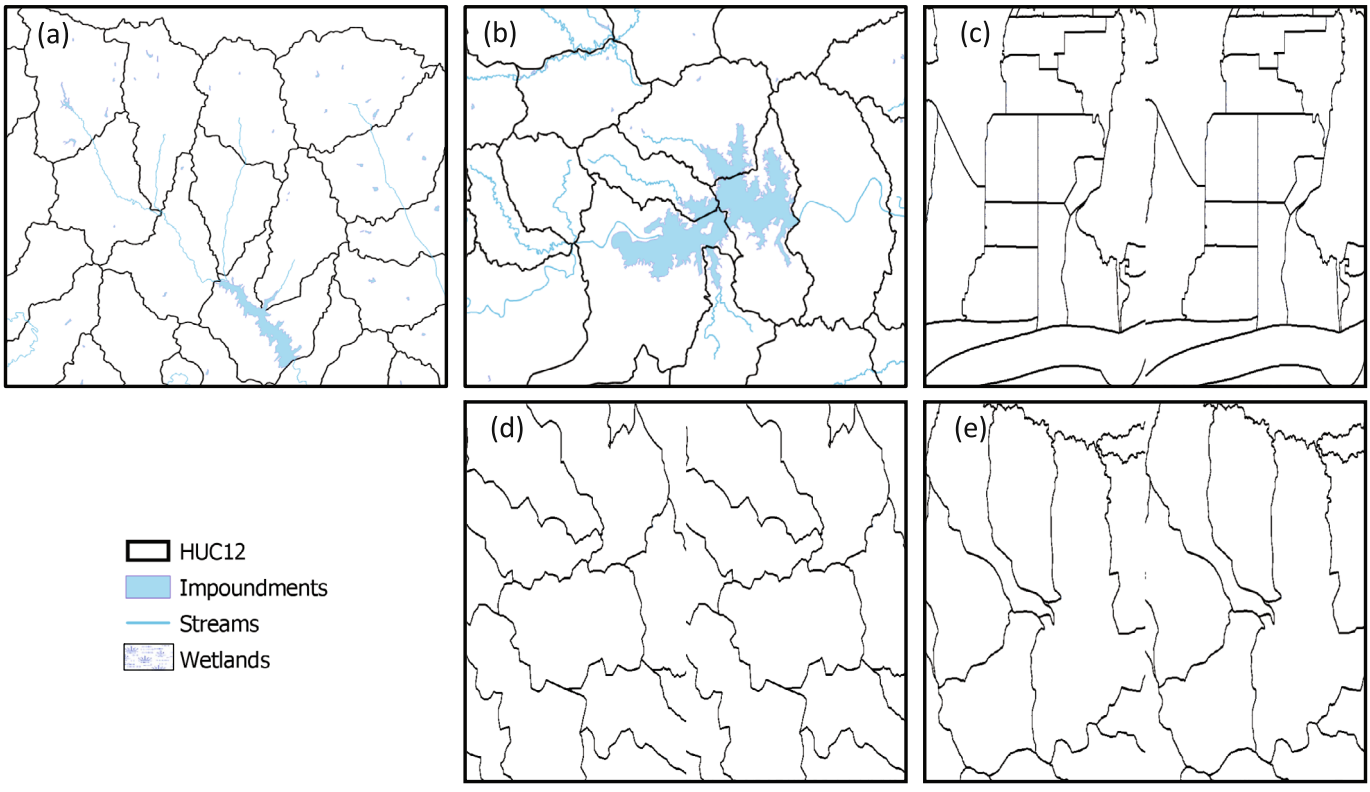


FIGURE 2. Example basin configurations, derived from NHDplus including: (a) Traditional connectivity, (b) reservoirs spanning multiple HUC12s, (c) entire HUC12 is wetlands, (d) playa lakes and no drainage network, (e) unconnected reservoirs. HUC, Hydrologic Unit Code; NHDplus, National Hydrography Dataset Plus.

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water bodies attached to the NHDPlus channel network are considered reservoirs, whereas non-connected structures are treated as ponds. Surface flow from each NHDPlus object (channel or reservoir) is directed to the next downstream channel or reservoir. Within each HUC12, NHDPlus connectivity is assumed to be correct, but connections between HUC12s are subject to additional data and constraints. For each HUC12, an outlet is added to which all surface flows are ultimately routed. To prevent infinite loops, groundwater contributions from channels and reservoirs cannot be directed to the shallow aquifer and are also directed to the outlet. If a municipal point source is designated, it also contributes directly to the outlet. The HUC12 outlet offers a conveniently described object from which to extract routed model predictions.

All non-headwater HUC12s have a single designated input object to receive upstream contributions. This object may be a stream segment or connected water body. There are two sources to describe the connectivity between HUC12s, the NHDPlus, and the 2015 WBD which was originally used to define the HUC12 boundaries. The WBD was found to contain a significant number of connectivity errors and was subject to manual correction during NAM development and previous efforts under the HAWQS (Yen, Daggupati, et al. 2016) program. Because the WBD was subjected to extensive manual correction, it was considered the most authoritative source. When WBD and NHDPlus disagree on HUC12 connectivity, a custom software program determines the most likely connecting channels and reservoirs in both the upstream and downstream HUC12s. On occasion, this leads to a skipped channel segment, particularly when NHDPlus channel segments cross-established WBD boundaries.

Point Sources

Point sources were derived primarily from the National Permit Discharge Elimination System (NPDES), which contains measured pollutant concentrations from permitted wastewater discharge facilities. These data were summarized by Maupin and Ivahnenko (2011) for 2002 and by Skinner and Maupin (2019) for 2012. These data were summarized by HUC12 and included as a direct addition to the outlet (Figure 3). Where data were lacking, U.S. Census population data (urban) and estimated pollutant loads per capita were used. Pollutant loads per

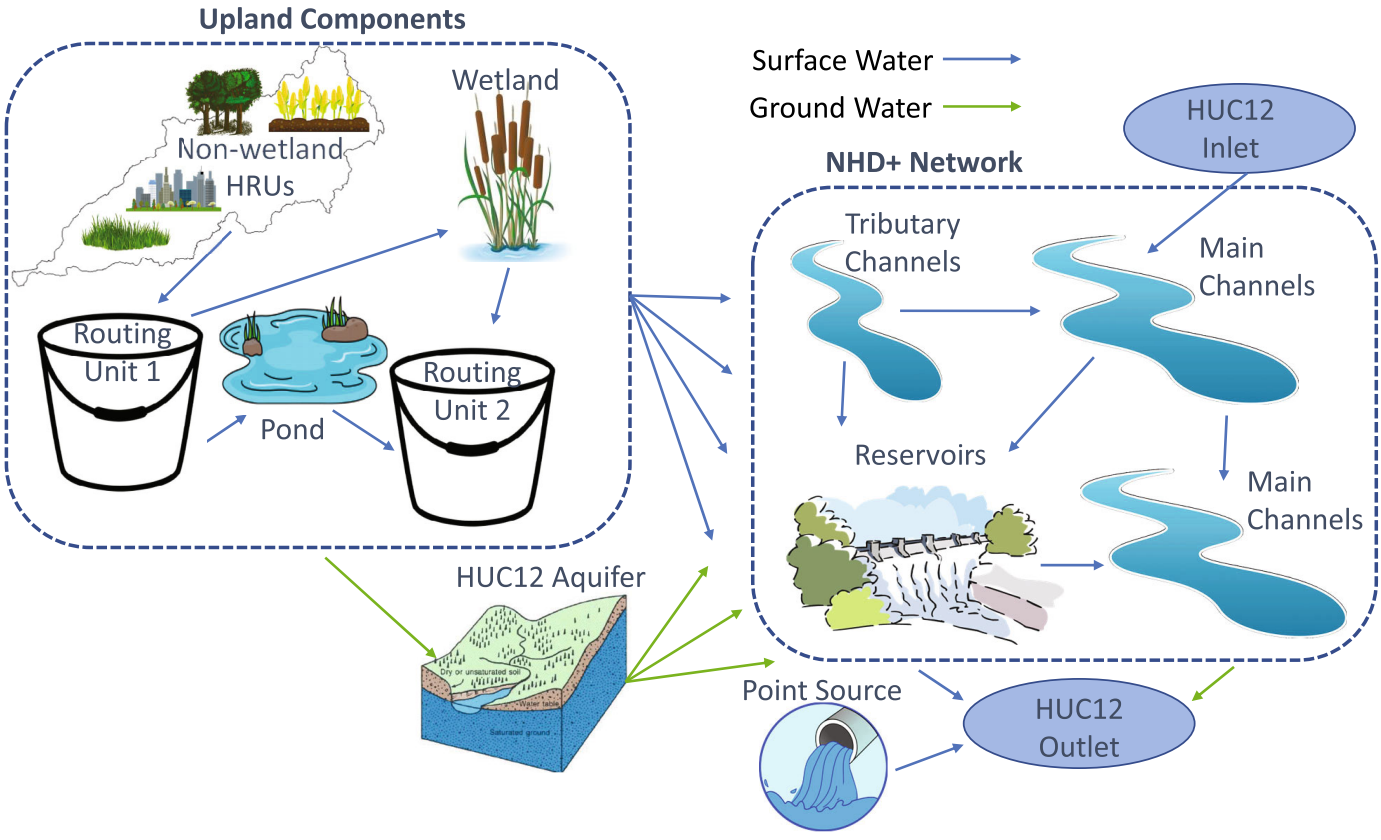


FIGURE 3. Object connectivity used in NAM. HRU, Hydrologic Response Unit.

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capita were derived from regional regression models where both NPDES and Census data at the HUC12 level were available. Average per capita loads averaged 4.14 kg/person/year for nitrogen and 0.5 kg/person/year for phosphorus. These estimates compare well with literature-reported values from Henze (2008) for nitrogen (0.73 to 5.5 kg/year/person) and phosphorus (0.37 to 1.01 kg/year/person). Point sources were included as annual loads in SWAT+. For 2002 and prior years, estimates for 2002 were used and after 2011, 2012 loads were used. Loads between 2002 and 2012 were linearly interpolated.

Water Use

Irrigation Sources. The source for irrigation water for each HRU was also defined in a probabilistic fashion using data from Dieter et al. (2018). They provided estimated surface and ground agricultural irrigation withdrawals by county, which were summarized by HUC8. Within each HUC8, the largest reservoirs and streams were identified as potential water sources. Surface water demand is taken preferentially from the largest reservoir, then the largest channel. HRUs identified to utilize groundwater irrigation were assumed to also contain the source water well. Deep aquifers underlying these HRUs were used as the primary source.

Municipal Usage. Municipal water usage comprises 12% of all water usage in the U.S. (Dieter et al. 2018). Total surface and groundwater withdrawals for municipal usage by county were taken from Dieter et al. (2018). These data were summarized to the HUC8 level and incorporated into each SWAT+ model. As with irrigation sources, municipal surface water demand is taken preferentially from the largest reservoir, then the largest channel. Municipal groundwater demand is taken from an unlimited deep aquifer source. We could have used shallow aquifer sources, but we would have had to identify which HUC12 aquifers to use without data. Furthermore, due to the variable capacity and distribution of these aquifers, we deemed them too uncertain for municipal supply.

File Structure and Execution

NAM was constructed using orders of magnitude more HRUs and channels than any previous SWAT watershed modeling effort, which presents a problem with storage, memory, and execution time. It is not practical nor desirable to build NAM as a single SWAT+ model. Because SWAT+ keeps all weather data for the entire simulation period and all state variables in system memory during model execution, we estimate that NAM would require 400GB of RAM for a 15-year simulation, with longer simulations requiring more. Therefore, we elected to break NAM into 2,109 individual HUC8 SWAT+ models. These models can be executed in a multi-threaded manner to drastically reduce computational time. The entire daily hydrograph including daily flow and constituent loads is saved to a file at the outlet of each HUC8 for each model. HUC8s that receive flow from upstream HUC8s are configured to accept a saved daily hydrograph file as an input. Individual HUC8 models are arranged in a hierarchical execution structure (Figure 4). If a HUC8 model is a headwater, that is if it does not receive flow from another HUC8, it has no dependencies and can be considered fully independent. If a HUC8 does receive flow from an upstream HUC8 then the downstream model requires an input hydrograph file from the upstream model. The downstream model can only be executed after the upstream model is finished. This approach allows the entire U.S. to be run much more quickly than would otherwise be possible, provided many CPU cores are available.

Most of the HUC8s in the CONUS are headwaters (61%) or 2nd execution order (16%). The last model in the execution order is at the outlet of the Mississippi River. To complete the entire set of 2,109 models, 31 execution levels are necessary, theoretically speeding up execution time over a single model by a factor of 68 (2,109/31) if unlimited parallel threads are possible.

In real-world applications, many factors impact model execution times. There is overhead involved in the moving of hydrograph files, delays in model execution between levels to prevent file locking issues, and bottlenecks in memory and disk access. For the purposes of testing, the entire NAM was executed on a single high-performance machine with differing levels of parallelism. The test CPU was an AMD Ryzen™ Threadripper™ 3990X with 64 physical cores (128 threads) with a base clock of 2.9 GHz, and a maximum boost clock of 4.3 GHz. The model was stored on a 2 Tb NVME drive array (RAID 0) and 128 Gb of memory was installed under Microsoft™ Windows 10 pro. The total execution time for a 15-year NAM simulation was 6,880 min using a single thread. Figure 5 illustrates active threads over time during the execution of NAM with differing allowable active threads. Despite having 64 physical cores, maximum performance was achieved with 32 threads (438 m) representing a 16-fold improvement in this example. This is likely due to non-CPU bottlenecks and variable CPU frequency under differing loads. Under lightly threaded loads, the 3990X CPU will

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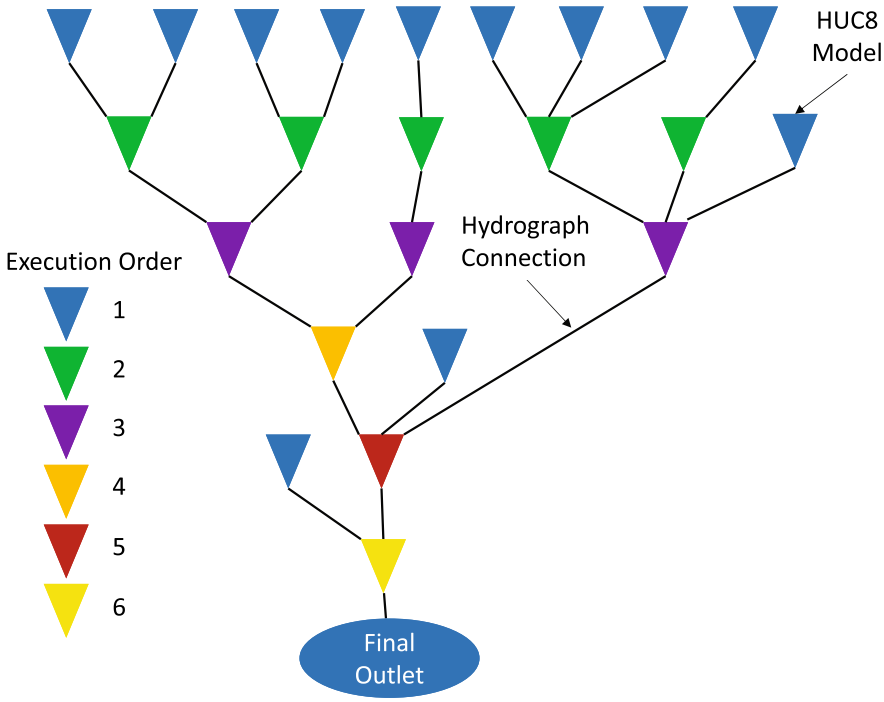


FIGURE 4. Conceptual NAM model execution hierarchy.

run at the boost frequency (4.3 GHz) as opposed to the base frequency (2.9 GHz) when heavily loaded.

An added benefit of the NAM hierarchical structure is the ability to run subsets of the model at any location. This allows tremendous multithreaded opportunities during model calibration. The calibration of model predictions that do not require routing between models like upland water balance (ET, runoff, percolation) or crop yields can be fully parallel, provided sufficient CPUs are available. Calibration for routed predictions (streamflow, nutrient loads, reservoir levels) need only run the models in the drainage of the entity where gage observations are available and are independent of all other models. Calibration at many locations simultaneously is possible if each gage lies in an independent watershed.

CASE STUDY

As an illustrative application of the NAM framework, we calibrated the average annual streamflow for a small subset of the model. We extracted and calibrated a single HUC8 model (Little River, 03110204) containing the Little River Experimental Watershed (Bosch et al. 2007), an LTAR site. The experimental watershed itself covers 334 km^2 area in the upper reaches of the Little River HUC8 (2,310 km^2), both of which lie within the Suwannee River Basin in south-central Georgia (Figure 6). The research watershed is subject to intensive long-term monitoring of both land management and associated instream water quality in nested watersheds. In this case study, we calibrated the entire HUC8 for streamflow using primarily nationally available data sources that are available for all HUC8s in the NAM framework. Additional model refinement to locally collected data within the Little River Experimental Watershed is certainly possible.

Streamflow data for the USGS site Little River near Adel, Ga (02318000), were obtained and summarized annually. Although there are other USGS gage data in the watershed, this one had the most complete records. The gage location was overlain with NHDPlus V2 to identify the unique stream segment in NAM that best represents the gage location. This gage lies directly below Reed Bingham Park Lake, a 151 Ha impoundment that is included in the model. The entire HUC8 (03110204) was executed in SWAT+ for the period 2003–2015 to obtain uncalibrated predictions of streamflow at the gage location. Because it is a single HUC8, multithreaded execution was not possible. The uncalibrated NAM underpredicted average flow by 33% (14.6 vs. 9.8 m^3/s).

The model was calibrated in two phases; a soft calibration of the upland water balance was performed, followed by a hard calibration to the stream gage. Soft data are defined as information on individual

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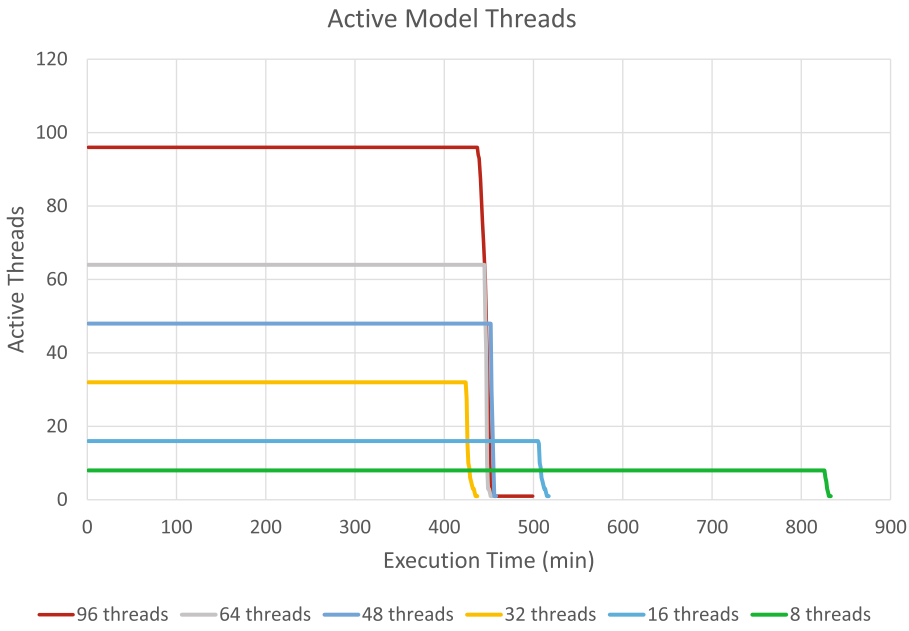


FIGURE 5. Active model threads over time during NAM execution on an AMD Ryzen™ Threadripper™ 3990X with 64 physical cores (128 threads) CPU.

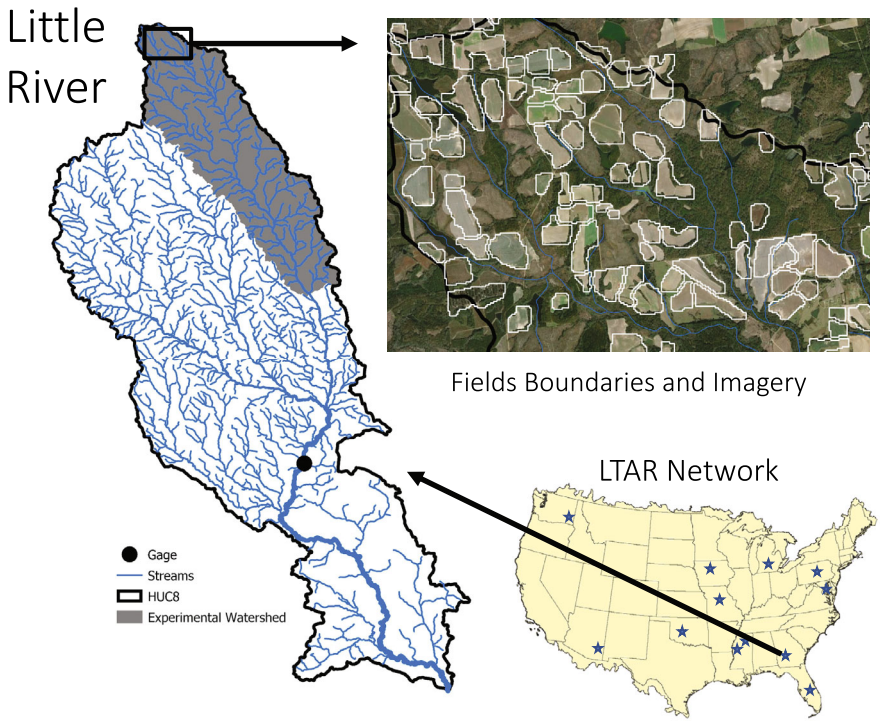


FIGURE 6. Little River HUC8 Watershed and Little River Experimental Watershed locations. LTAR, Long-Term Agroecosystem Research.

processes within a budget that may not be directly measured within the study area, maybe an average annual estimate and may entail considerable uncertainty. Hard data are more rigorous, representing observational data directly comparable with some element in the model, such as sediment load within a particular simulated stream reach. The use of hard and soft calibration strategies has been employed in several SWAT efforts (Yen, White, et al. 2016; Chawanda et al. 2020). Upland water balance components (precipitation, surface runoff, ET, total water yield, baseflow) were derived from Baffaut

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TABLE 1. Parameter adjustments made in SWAT+ for Little River HUC8 and Experimental Watershed.

Parameter	Parameter description	Type of change	Parameter value
Modified during soft calibration			
lat_ttime	Exponential of the lateral flow travel time	Absolute change	0.295
esco	Soil evaporation compensation factor	Absolute change	0.05
cn3_swf	Soil water at curve number III (0 = field capacity; 0.99 = saturation)	Absolute change	−0.935
perco	Percolation coefficient	Absolute change	0.1
harg_pet	Radiation coefficient used in Hargreaves equation	Absolute change	−0.00036
latq_co	Lateral flow travel time coefficient	Absolute change	0.518
Modified during hard calibration			
revap_co	Fraction of pet to calculate revap	Fractional change	0.05
revap_min	Threshold depth to water in shallow aquifer from surface to allow revap	Absolute value	3.25
cn3_swf	Soil water at curve number III (0 = field capacity; 0.99 = saturation)	Absolute change	0.05
awc	Available water capacity of soil layer	Fractional change	0.05
cn2	Condition II curve number	Absolute change	−0.25
flo_min	Minimum aquifer storage to allow return flow	Absolute value	3.75
esco	Soil evaporation compensation factor	Absolute change	0.02

Note: SWAT, Soil and Water Assessment Tool.

et al. (2020) and supplemented by Reitz et al. (2017). Parameters were adjusted to match water balance components expressed as a fraction of average annual precipitation. The soft calibration parameters are given in Table 1. Soft calibration of NAM reduced the overprediction of stream flow at the gage to 8.2% (14.6 vs. 15.8 m^3/s).

Additional parameter adjustments were made during the hard calibration to further refine average annual model predictions at the stream gage with the objective to minimize relative error in total flow. Iterative calibration adjustments were halted when the average annual relative error fell below 1% (14.6 vs. 14.7 m^3/s). Annual observed and predicted streamflow at each calibration stage is given in Figure 7.

DISCUSSION AND CONCLUSIONS

National Agroecosystems Model was primarily developed to support USDA modeling efforts within LTAR and CEAP. To support these efforts, NAM must be able to simulate a variety of agricultural scenarios including: "business as usual" and "aspirational" cropping systems defined by LTAR; future

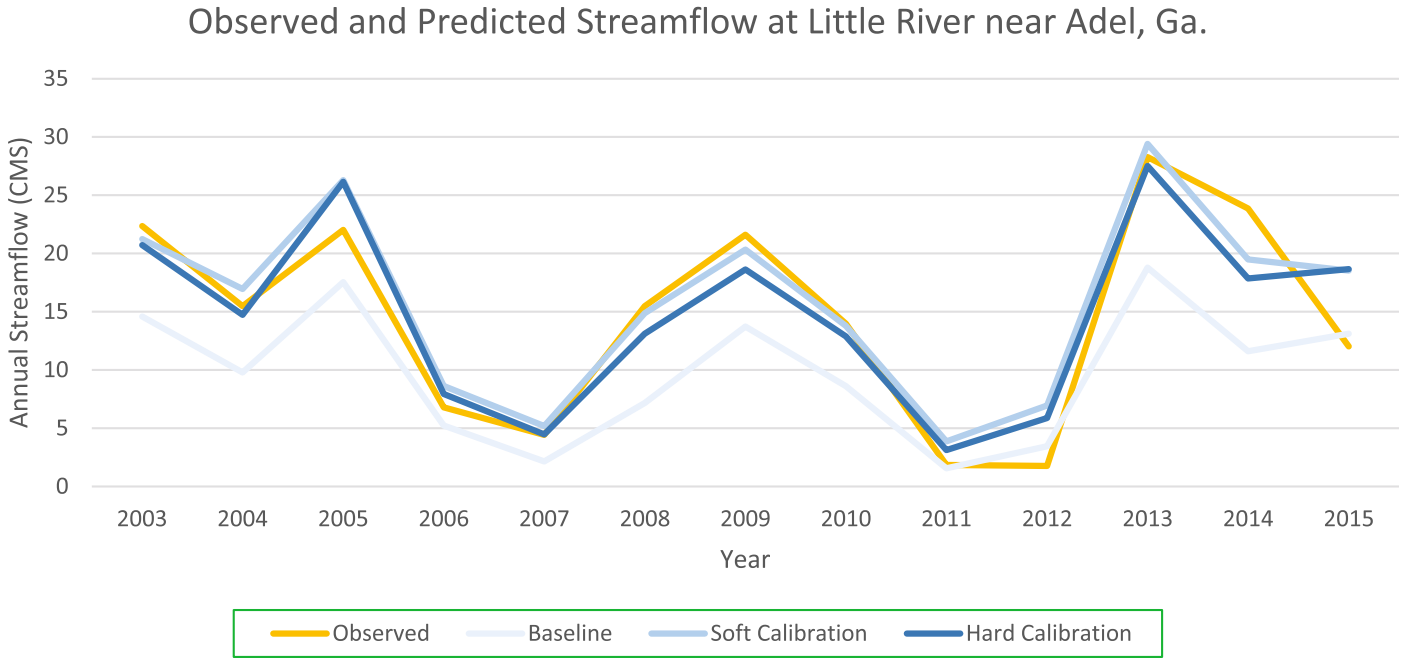


FIGURE 7. Annual observed and predicted streamflow at each calibration stage in cubic meters per second (CMS).

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levels of conservation adoption to inform national agricultural policy for CEAP; climate change impact on agriculture and water supply; effects of conservation programs on wildlife; and simulations to drive field-scale conservation planning tools. These needs require a very detailed model with a large spatial extent, a combination that is inherently difficult. The scope of this task required the development of custom software and the adoption of enterprise database tools and hardware. The ultimate goal of NAM is to support projects beyond LTAR and CEAP, with either predefined and executed scenarios or raw models suitable for modification, refinement, and adaptation to project-specific objectives.

Because NAM is constructed with only publicly available data, it can be shared to support research with other government agencies, universities, and others in the public domain. Being constructed as 2,121 HUC8 models allows it to be easily subset for a user's area of interest. Various portions of the NAM model are currently being utilized in collaborative efforts in the Mississippi River Basin and to power AgOPT (https://realtimeceap.brc.tamus.edu/), which aids producers in the scheduling the application of pesticides and fertilizers.

The NAM framework is intended to be versioned as new data are ingested or new calibrations are applied. It is continually updated with new data and improved methods. With advances in emerging technologies (satellites, drones, artificial intelligence, faster computers), new data sources are published at an ever-increasing pace. Most recently, 30-m resolution irrigation by year (Xie and Lark 2021) and tile drainage (Valayamkunnath et al. 2020) have been publicly released and are being formatted for ingestion in NAM. NAM is being calibrated using soft data to define reasonable watershed processes, i.e., water balances, nutrient cycling, and crop yield. Hard calibration is also underway to further adjust NAM to match specific sediment and nutrient loads. NAM is not intended to be a turnkey highly accurate model applicable to every situation and scale. Ultimately, we seek to provide users with a documented, consistent, and reasonable base model, which can be refined to address their particular project objectives. Users may need to enhance model inputs and peruse further calibration to reduce and define uncertainties in their application. More general releases of NAM are scheduled for 2022 and subsequent years.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

AUTHOR CONTRIBUTIONS

Michael J. White: Conceptualization; data curation; formal analysis; funding acquisition; investigation; methodology; project administration; resources; software; supervision; validation; visualization; writing – original draft. Jeffrey G. Arnold: Conceptualization; data curation; formal analysis; funding acquisition; investigation; methodology; project administration; resources; software; supervision; validation; visualization; writing – review and editing. Katrin Bieger: Conceptualization; data curation; formal analysis; investigation; methodology; project administration; supervision; validation; visualization; writing – review and editing. Peter M. Allen: Conceptualization; formal analysis; investigation; methodology; supervision; visualization; writing – review and editing. Jungang Gao: Data curation; formal analysis; investigation; methodology; software; validation; visualization; writing – review and editing. Natalja Čerkasova: Data curation; formal analysis; investigation; methodology; software; validation; visualization; writing – review and editing. Marilyn Gambone: Data curation; investigation; software; visualization; writing – review and editing. Seonggyu Park: Software; visualization; writing – review and editing. David D. Bosch: Validation; writing – review and editing. Haw Yen: Data curation; investigation; methodology; software; validation; writing – review and editing. Javier M. Osorio: Formal analysis; investigation; software; validation; visualization; writing – review and editing.

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